Hybrid Information Retrieval for Tip-of-the-Tongue Dataset

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***Abstract*—In this research, the impact of retrieval techniques such as lexical, dense, and hybrid models is investigated on the Tip-of-the-Tongue (ToT) dataset. The BM25 model is used for lexical retrieval. SBERT and BGE-M3 are used as dense retrieval models. In addition, hybrid approaches combining lexical and dense models are evaluated. To analyze model performance across different query groups. All models are evaluated using standard information retrieval metrics, including Recall@K, Mean Reciprocal Rank (MRR), and NDCG@K.**

***Keywords*—BM25, SBERT, BGE-M3, MonoT5, Hybrid Retrieval, Recall@K, MRR, NDCG@K**

I. Introduction

The Information Retrieval (IR) systems strive to extract relevant information from vast document repositories in response to user queries. Nevertheless, user queries may be ambiguous and vague. One instance of query ambiguity is the Tip-of-the-Tongue (ToT) phenomenon when users recall the target information partly but fail to specify their queries accurately. As a result, vocabulary mismatches become common and hamper the effectiveness of lexical retrieval models, such as BM25.

A series of recent works [1] shows that dense embedding models, like SBERT, can handle semantically ambiguous queries by leveraging the contextual semantic similarity of embeddings. Moreover, advanced dense retrieval models, including BGE-M3 [3], combine the lexical and semantic retrieval signals into one model. Furthermore, recent research [2] has shown that powerful neural reranking models, such as MonoT5, can enhance retrieval performance in hybrid retrieval systems.

In our experiments, we assess the performance of lexical, dense, hybrid, and neural reranking retrieval models on the TREC 2024 Tip-of-the-Tongue retrieval task. We consider the following retrieval systems: BM25, SBERT, BGE-M3, MonoT5, and BGE-Reranker. The retrieval systems will be evaluated by Recall@k, MRR, and NDCG@k measures.

In order to measure the degree of ambiguity in a query, we model query vagueness using the average IDF score of terms contained in the query, such that low scores imply general queries while high scores imply specific queries. Our assumption is that semantic retrieval models perform better than lexical retrieval techniques due to contextual similarity.

II. Literature Review

Conventional information retrieval systems largely depend on lexical retrieval approaches like BM25. Study [4] notes that BM25 evaluates documents' relevance using frequency statistics of the terms contained in them. While BM25 is good at retrieving lexically similar documents, it usually lacks the ability to retrieve semantically similar documents when users query for them. This is highly significant in the case of the Tip-of-the-Tongue (ToT) retrieval problem since user queries are usually incomplete and semantically ambiguous.

As a solution to this problem, there have been some dense retrieval models developed using neural embeddings. For instance, SBERT [1] enhances semantic retrieval using contextualized sentence embeddings, whereas recent neural dense models, such as BGE-M3 [3], combine both semantic and lexical retrieval cues into one framework. SBERT and BGE-M3 are chosen in this research due to their excellent abilities in semantic retrieval.

Apart from dense retrieval, current research findings also show that hybrid retrieval systems utilizing both lexical and semantic signals in retrieving documents achieve significant enhancements in retrieval performance [2]. Accordingly, hybrid retrieval pipelines incorporating BM25 and dense retrieval scores through weighted interpolation will be examined in this research. Moreover, the use of MonoT5 [2] as a neural reranking approach will be considered to determine whether interactive reranking would add more value to the hybrid retrieval process.

Regarding the retrieval mechanism, BM25 retrieval requires minimal preprocessing steps such as lowercase conversion, punctuation elimination, and stopwords removal. Nevertheless, for dense retrieval methods, textual input is not preprocessed as it heavily depends on the semantic nature of texts.

III. Research Questions and Methodology

A. Research Questions

RQ1 (Lexical-Semantic Gap): Can zero-shot dense retrieval models such as SBERT and BGE-M3 outperform BM25 in capturing the semantic meaning of vague Tip-of-the-Tongue (ToT) queries?

RQ2 (Hybrid Synergy): Can weighted interpolation of lexical and semantic retrieval scores provide statistically significant improvements over standalone lexical and dense retrieval approaches?

RQ3 (Architecture Comparison): Is linear hybrid fusion more effective and computationally efficient for large candidate-space ranking compared to neural reranking architectures such as MonoT5 and BGE-Reranker?

B. Proposed Architecture

The proposed retrieval framework follows a modular multi-stage retrieval pipeline designed for large-scale retrieval over approximately 3.1 million documents.

Step 1 – Candidate Selection: BM25 is used as the initial lexical retriever to obtain the top-10,000 candidate documents for each query. These candidate documents form the retrieval space for subsequent neural retrieval models.

Step 2 – Neural Enhancement: Two different neural retrieval strategies are investigated in parallel:

- **Hybrid Retrieval:** Dense retrieval scores produced by SBERT and BGE-M3 are combined with BM25 scores using min-max normalization and weighted linear interpolation with coefficient λ.
- **Neural Reranking:** MonoT5 and BGE-Reranker models directly rerank the BM25 candidate documents through query-document interaction without explicit score fusion. Neural reranking experiments required substantially higher computational cost compared to lexical retrieval baselines.

Step 3 – Comparative Evaluation: All retrieval systems are evaluated using nDCG@10, nDCG@1000, Recall@10, Recall@1000, and RR@10 metrics in order to assess retrieval effectiveness at both early and deep ranking levels.

C. Experiment Tracking and Statistical Significance

Hyperparameter sweeps and experimental tracking are conducted using Weights & Biases (WANDB). In particular, λ interpolation coefficients are systematically evaluated for hybrid retrieval models. Statistical significance tests based on paired t-tests ($p < 0.05$) are performed to compare the proposed approaches against the BM25 baseline.

IV. Data Overview

In this study, the TREC 2024 Tip-of-the-Tongue dataset is used for an information retrieval task. It contains 3 parts: queries, qrels, and a document corpus. The aim is to find correct document from corpus based on queries. However, these queries sometimes do not contain exact keywords; instead, they are unclear and vague. It has three splits: train, dev1, and dev2. For this study, the train split is used for training, dev1 is used for validation, and dev2 is used for testing. The queries split has 150 queries, divided into two columns: query_id and query. The query_id is the unique identifier, and query is the query string. In addition, the qrels split has four columns: query_id, iteration, doc_id, and relevance. The relevance column contains relevance label, and all queries in dataset are relevant to one document. Also, corpus has 3,185,450 documents in five columns: doc_id, title, wikidata_id, text, and sections. The number of queries is small, but the corpus is very large. This is a challenging scenario in which model has to select the right document from large corpus based on vague description. The dataset is clean and doesn't have any missing values or duplicate records in data quality part. Hence, we don't have to perform any data cleaning.

V. Exploratory Data Analysis

TABLE I. Summary Statistics of Queries

Split	Number of Queries	Number of Qrels	Avg. Query Length	Median Query Length
Train	150	150	147.38	127.0
Validation	150	150	139.33	118.5
Test	150	150	144.3	125.5

The number of queries and qrels is balanced for all splits. The average length of queries is relatively high, suggesting that queries are descriptive rather than keyword-based. The average and median lengths of queries are also close.

Semantic similarity between queries was computed using a pre-trained SBERT model (all-MiniLM-L6-v2). The average cosine similarity values were moderate and close across all splits: 0.3577 for train, 0.3717 for validation, and 0.3705 for test. This suggests that the dataset is relatively consistent across splits, while still containing semantically diverse queries. Moreover, most queries have medium length, although some longer queries exist. These observations indicate that ToT queries are generally descriptive rather than purely keyword-based, which may give semantic retrieval models such as SBERT, BGE-M3, and MonoT5 an advantage over purely lexical models such as BM25.

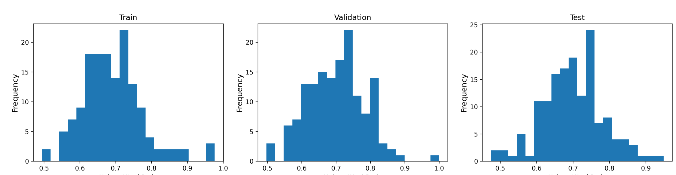

Fig. 1. Distribution of unique words across train, validation, and test splits

The relatively high unique-word ratio suggests that many queries are descriptive and lexically varied. This may make exact lexical matching harder and motivates the use of semantic retrieval models in addition to lexical baselines.

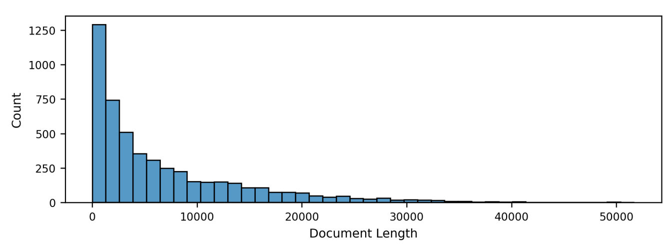

Fig. 2. Distribution of document lengths in the corpus.

Due to the large corpus size (3M+ documents), we analyze a sampled subset for efficiency. The distribution of corpus length is right-skewed. Most of the documents are of relatively short length, while some of the documents are of extremely long length. There are some outliers in the data.

VI. Challenges of Tip-of-the-Tongue Retrieval Task and Related Works

In comparison with conventional ad-hoc retrieval queries, the TREC 2024 Tip-of-the-Tongue (ToT) retrieval task is intrinsically more difficult since the user queries are frequently vague, uncompleted, and have semantic ambiguity. Thus, state-of-the-art neural retrieval models obtain only moderate performance scores for their retrieval effectiveness. As can be seen from the TREC ToT 2024 benchmarks [5], the performance score of the BM25 model (k1=1, b=1) is 0.0657 in terms of NDCG@10; the NDCG@10 score of the dense retrieval model based on SBERT is 0.1040, while its NDCG@1000 is 0.1665. DSGT at TREC ToT 2025 [6] studied further hybrid retrieval pipelines that combine sparse retrieval, dense retrieval, LLM-based retrieval, and reranking techniques. The researchers found out that hybrid lexical-semantic retrieval systems significantly outperform lexical-only baselines.

VII. Experimental Setup and Hyperparameter Tuning

All experiments were conducted using fixed random seeds and logged through Weights & Biases in order to ensure experiment reproducibility and transparent hyperparameter tracking. The experimental results indicate that combining lexical and semantic retrieval signals produces more stable retrieval performance than using either approach independently. In particular, hybrid retrieval consistently improved nDCG@10 and deep-ranking effectiveness across multiple settings.

A. BM25 Candidate Size Analysis

To determine the candidate size, multiple depths of retrieval of BM25 were experimentally evaluated. As shown in Table II. retrieval effectiveness saturated after approximately 1,000 candidates, while candidate recall continued to increase substantially for larger retrieval depths. Therefore, a candidate size of 10,000 documents was selected in order to maximize semantic coverage for downstream dense retrieval and neural reranking architectures.

TABLE II
BM25 Candidate Size Analysis

Candidate Size	Candidate Recall	nDCG@10	nDCG@1000
50	0.1867	0.0805	0.0967
100	0.2333	0.0805	0.1041
200	0.3067	0.0805	0.1143
500	0.3667	0.0805	0.1214
1000	0.4533	0.0805	0.1306
2000	0.5400	0.0805	0.1306
5000	0.6067	0.0805	0.1306
10000	0.6867	0.0805	0.1306
20000	0.7667	0.0805	0.1306

B. BM25 Parameter Tuning

After selecting the candidate retrieval depth, BM25 hyperparameter tuning was conducted using Weights & Biases. The parameters k_1 and b were swept over different values, and the runs were compared using Recall@10, RR@10, nDCG@10, nDCG@100, nDCG@1000, and candidate recall. Table III shows that the best configuration was obtained with $k_1 = 0.9$ and $b = 0.9$, which achieved the highest candidate recall while also maintaining strong ranking effectiveness across multiple cutoff levels.

TABLE III
BM25 Parameter Tuning Results

k_1	b	Recall@10	RR@10	nDCG@10	nDCG@100	nDCG@1000	Candidate Recall
0.9	0.90	0.1067	0.0719	0.0805	0.1041	0.1306	0.7667
1.5	0.75	0.1000	0.0663	0.0747	0.1004	0.1258	0.7600
1.2	0.75	0.1000	0.0680	0.0761	0.0991	0.1273	0.7600
0.9	0.75	0.1067	0.0721	0.0808	0.1054	0.1281	0.7600
1.2	0.90	0.1067	0.0706	0.0792	0.1036	0.1286	0.7467
1.5	0.90	0.1067	0.0678	0.0767	0.1003	0.1273	0.7400
0.9	0.40	0.1000	0.0560	0.0669	0.0822	0.1010	0.7067
1.2	0.40	0.1000	0.0532	0.0648	0.0823	0.1007	0.7067
1.5	0.40	0.1067	0.0527	0.0656	0.0815	0.1007	0.7000

C. Hybrid Retrieval Weight Optimization

After choosing the settings for the BM25 model (with $k_1 = 0.9$, $b = 0.9$) and a candidate list size of 10,000, hybrid search was evaluated using the SBERT and BGE-M3 dense retrieval models.

The hybrid search approach used BM25 and dense retrieval scores normalized using min-max normalization with weighted linear interpolation applied as follows:

$$Score(q, d) = (1 - \lambda) \cdot BM25(q, d) + \lambda \cdot Dense(q, d)$$

where $\lambda \in [0, 1]$ determines the weighting of the signals provided by both lexical and semantic retrievals.

If $\lambda = 0$, then the method would be the one for BM25 search, while $\lambda = 1$ would stand for the pure dense retrieval model.

TABLE IV
Hybrid Retrieval Weight Optimization Results

Model	λ	nDCG@10	nDCG@100	nDCG@1000	RR@10	Recall@10
SBERT	0.0	0.0805	0.1041	0.1306	0.0719	0.1067
SBERT	0.1	0.0839	0.1144	0.1398	0.0765	0.1067
SBERT	0.2	0.0885	0.1217	0.1476	0.0786	0.1200
SBERT	0.3	0.1015	0.1313	0.1608	0.0878	0.1467
SBERT	0.4	0.1157	0.1484	0.1768	0.1018	0.1600
SBERT	0.5	0.1375	0.1670	0.1988	0.1223	0.1867
SBERT	0.6	0.1505	0.1834	0.2136	0.1370	0.1933
SBERT	**0.7**	**0.1530**	**0.1896**	**0.2175**	**0.1389**	**0.2000**
SBERT	0.8	0.1366	0.1807	0.2061	0.1251	0.1733
SBERT	0.9	0.1133	0.1516	0.1819	0.0983	0.1600
SBERT	1.0	0.0985	0.1314	0.1635	0.0842	0.1467
BGE-M3	0.0	0.0805	0.1041	0.1306	0.0719	0.1067
BGE-M3	0.1	0.0839	0.1109	0.1384	0.0765	0.1067
BGE-M3	0.2	0.0879	0.1178	0.1445	0.0780	0.1200
BGE-M3	0.3	0.0962	0.1271	0.1536	0.0850	0.1333
BGE-M3	0.4	0.1092	0.1408	0.1665	0.0955	0.1533
BGE-M3	0.5	0.1196	0.1510	0.1790	0.1046	0.1667
BGE-M3	0.6	0.1377	0.1679	0.1951	0.1223	0.1867
BGE-M3	**0.7**	**0.1477**	**0.1816**	**0.2047**	**0.1335**	**0.1933**
BGE-M3	0.8	0.1283	0.1657	0.1871	0.1139	0.1733
BGE-M3	0.9	0.1069	0.1360	0.1608	0.0881	0.1667
BGE-M3	1.0	0.0775	0.1045	0.1306	0.0625	0.1267

D. Neural Reranking Experiments

Aside from dense hybrid search, interaction-based neural rerankers were tested using BGE-Reranker and MonoT5. An interpolation approach was also used in testing the difference between reranker performance alone at $\lambda = 1$ and hybrid reranking with BM25.

TABLE V
NEURAL RERANKING WEIGHT OPTIMIZATION RESULTS

Model	λ	nDCG@10	nDCG@100	nDCG@1000	RR@10	Recall@10
BGE-RR	0.0	0.0805	0.1041	0.1306	0.0719	0.1067
BGE-RR	0.1	0.0856	0.1121	0.1394	0.0786	0.1067
BGE-RR	0.2	0.0881	0.1219	0.1492	0.0819	0.1067
BGE-RR	0.3	0.1048	0.1395	0.1642	0.0961	0.1333
BGE-RR	0.4	0.1178	0.1534	0.1807	0.1089	0.1467
BGE-RR	0.5	0.1217	0.1586	0.1864	0.1098	0.1600
BGE-RR	**0.6**	**0.1362**	**0.1713**	**0.1965**	**0.1189**	**0.1933**
BGE-RR	0.7	0.1284	0.1600	0.1855	0.1029	0.2133
BGE-RR	0.8	0.1179	0.1528	0.1770	0.0950	0.1933
BGE-RR	0.9	0.0904	0.1329	0.1531	0.0731	0.1467
BGE-RR	1.0	0.0711	0.1033	0.1272	0.0523	0.1333
MonoT5	0.0	0.0805	0.1041	0.1306	0.0719	0.1067
MonoT5	**0.1**	**0.0809**	**0.1044**	**0.1321**	**0.0725**	**0.1067**
MonoT5	0.2	0.0797	0.1030	0.1292	0.0710	0.1067
MonoT5	0.3	0.0768	0.0997	0.1263	0.0671	0.1067
MonoT5	0.4	0.0660	0.0914	0.1154	0.0573	0.0933
MonoT5	0.5	0.0543	0.0745	0.0959	0.0437	0.0867
MonoT5	0.6	0.0399	0.0557	0.0765	0.0317	0.0667
MonoT5	0.7	0.0351	0.0488	0.0639	0.0295	0.0533
MonoT5	0.8	0.0307	0.0403	0.0525	0.0238	0.0533
MonoT5	0.9	0.0215	0.0316	0.0428	0.0157	0.0400
MonoT5	1.0	0.0117	0.0191	0.0290	0.0089	0.0200

Out of all the experimental setups, the highest-performing hybrid approach for information retrieval was that of SBERT with $\lambda = 0.7$, attaining an nDCG@10 value of 0.1530 and nDCG@1000 value of 0.2175. Also, BGE-M3 performed exceptionally well in hybrid retrievals, with the best setup occurring when $\lambda = 0.7$. From these results, it is clear that hybrid retrievals involving dense and lexical signals perform significantly better than dense retrievals alone.

In neural reranking experiments, BGE-Reranker performed much better than MonoT5. The most optimal configuration for BGE-Reranker yielded an nDCG@10 value of 0.1362 with $\lambda = 0.6$ while MonoT5 had 0.0809 when $\lambda = 0.1$. This poor performance of MonoT5 may be due to the huge candidate set coupled with the vague nature of Tip-of-the-Tongue queries.

E. Statistical Significance Analysis

TABLE VI
STATISTICAL SIGNIFICANCE ANALYSIS ON NDCG@10

Comparison	Mean A	Mean B	Diff (%)	p-value	Sig.
Base SBERT vs Hybrid SBERT	0.0985	0.1530	55.33	0.1417	False
Base BGE-M3 vs Hybrid BGE-M3	0.0775	0.1477	90.56	0.0311	True
Base Reranker vs Hybrid Reranker	0.0711	0.1362	91.60	0.0350	True
Base MonoT5 vs Hybrid MonoT5	0.0117	0.0809	588.98	0.0018	True
BM25 vs Hybrid Reranker	0.0805	0.1362	69.25	0.0869	False
BM25 vs Hybrid MonoT5	0.0805	0.0809	0.57	0.9872	False
BM25 vs Hybrid SBERT	0.0805	0.1530	90.20	0.0382	True
BM25 vs Hybrid BGE-M3	0.0805	0.1477	83.50	0.0527	False
Hybrid BGE-M3 vs Hybrid MonoT5	0.1477	0.0809	-45.19	0.0015	True
Hybrid Reranker vs Hybrid MonoT5	0.1362	0.0809	-40.58	0.0002	True
Hybrid SBERT vs Hybrid MonoT5	0.1530	0.0809	-47.12	0.0005	True
Hybrid SBERT vs Hybrid BGE-M3	0.1530	0.1477	-3.52	0.7376	False

Table VI. shows the statistical significance testing of the results based on the metric nDCG@10. From the results, it is evident that the hybrid search algorithms exhibit better performance compared to the baseline methods. For instance, there was a notable improvement of 90.20% for Hybrid SBERT compared to the baseline BM25 approach with ($p < 0.05$). The same observation can be seen from the performance of the Hybrid BGE-M3 and the hybrid reranking systems.

Out of all the systems tested, the best retrieval performance was observed from Hybrid SBERT. Nonetheless, the difference in performance between the Hybrid SBERT and Hybrid BGE-M3 models was not statistically significant ($p = 0.7376$).

Finally, the monoT5 retrieval algorithm exhibited poor performance compared to other hybrid models. This could be attributed to the large size of the candidate list used in the experiment and the vague characteristics of Tip-of-the-Tongue search queries.

VIII. FUTURE WORKS

Although the existing retrieval pipeline works perfectly well, the last phase of the current research will be dedicated to further fine-tuning to eliminate the performance problems identified during our preliminary tests.

Based on our preliminary findings, the neural reranking models – MonoT5 and BGE-Reranker – show weaker performance compared to the hybrid fusion approach. This is most likely associated with the zero-shot transfer of those models on a domain-specific dataset. Thus, we will concentrate on the supervised fine-tuning of MonoT5 and BGE-Reranker on a subset of TREC ToT dedicated to the task at hand. As a result, we will make sure that the weights of those models are adapted to recognize the subtle semantic features of tip-of-the-tongue queries. Specifically, we will experiment with the following two hyper-parameters:

Depth of Re-ranking (k): We will experiment with different values of k (such as $k \in 50, 100, 250$) and see if the current drop in performance is associated with the high level of noise among candidates for re-ranking.

Maximum Input Length: For SBERT and BGE-M3, we will experiment with the maximum input sequence length in order to preserve full descriptive queries.

Every subsequent step of fine-tuning will be recorded using Weights & Biases (WANDB).

Finally, during the last stage, the highest performing model from all fine-tuned models will be identified using the nDCG@10 metric. Top 100 outputs of the chosen model will then be ranked using a Large Language Model (LLM), namely Gemma-2-9B, in the listwise re-ranking procedure, aiming to settle any semantic ambiguity within Tip-of-the-Tongue questions using the model's high-contextual capacity.

IX. COMPUTATIONAL LIMITATIONS

Because of computational constraints and the relatively large amount of data in the TREC ToT dataset (3.1 million documents), the dense retriever and neural re-ranking and dense were done using only a subset of candidates and not the entire search space. This subset was chosen by using BM25.

Besides, large-scale re-ranking using the language model could not be implemented entirely within this intermediate period owing to the highly computationally complex and memory-intensive nature of interaction models. This is elaborated further in the future works section. It is worth noting that this limitation led to a phased approach to utilize the language model for re-ranking where the top-100 documents are re-ranked by language models such as Gemma-2-9B instead of all candidate documents.

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Code Availability: The implementation of this project is available at:

https://github.com/Gozde123/trec-tot-retrieval

W&B Availability: The experiment logs and reports are available at Interim 1 W&B Report and Interim 2 W&B Report.